

Question 1: Heuristic design and admissibility

Steps

1. $h(s)$ = number of positions where s differs from the goal
2. Each mismatched position must be changed at least once, and one action changes only one position
3. Since every step costs 1, this is a lower bound on the true remaining cost; therefore it is admissible

Scrabble

1. $h(s)$ = sum of the Scrabble values of the goal letters at all mismatched positions
2. Every mismatched position must eventually receive its goal letter, so each corresponding replacement cost is unavoidable
3. Any intermediate substitutions only add nonnegative cost, so this sum cannot exceed the true optimal remaining cost; therefore it is admissible.

Frequency

1. $h(s) = 0$ at the goal; otherwise $h(s) = \max(\text{number of mismatches}, 1 + \text{rarity}(\text{goal}))$.
2. At least one action is required per mismatch, and every action costs at least 1. Also the final action into the goal necessarily costs $1 + \text{rarity}(\text{goal})$
3. Both terms are valid lower bounds, so their maximum is also a lower bound and cannot overestimate the true remaining cost; therefore it is admissible

Question 2: Does the heuristic actually help?

Word pair	Cost	UCS cost	UCS expanded	A* cost	A* expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.8554	808	38.8554	688	14.85%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402	1084	38.2402	427	60.61%

1. Did UCS and A* always return the same optimal cost? If not, identify and investigate the discrepancy.

Yes. For all six experiments, UCS and A* returned identical solution costs, so there was no discrepancy to investigate.

2. For each word pair, under which cost measure did A* produce the largest percentage reduction in expanded nodes?

WHY → ASK: Scrabble, with a 98.30% reduction.

BEAR → DUCK: Scrabble, with a 99.80% reduction.

3. For each word pair, under which cost measure was the heuristic least informative?

WHY → ASK: Frequency, with only a 14.85% reduction.

BEAR → DUCK: Frequency, with a 60.61% reduction.

4. Explain the differences using the definitions of your heuristics, not runtime alone.

Steps:

- The heuristic counts mismatched positions.

- Since every mismatch requires at least one action, it gives a strong direct estimate of the remaining number of steps.

Scrabble:

- The heuristic sums the Scrabble values of the goal letters that still must be inserted.
- This captures both the number of mismatches and their different costs, making it more informative than simply counting mismatches in these experiments.

Frequency:

- The heuristic uses $\max(\text{number of mismatches}, 1 + \text{rarity}(\text{goal}))$.
- It does not account for the rarity costs of the intermediate words that may be required along the path, so it distinguishes states much less precisely.
- This explains why A* reduced expansions far less under frequency cost.

Question 3: Reversibility and direction-dependent costs

Steps:

1. Yes.
2. Every legal action is reversible, and every action costs 1 in either direction. Reversing any path therefore preserves its total cost, so the optimal cost is the same in both directions.
3. Example: $\text{dog} \rightarrow \text{cog}$ costs 1 and $\text{cog} \rightarrow \text{dog}$ costs 1.

Scrabble:

1. No.
2. Actions are reversible, but cost depends on the replacement letter, so reversing an action can change its cost.
3. Counterexample:
 - $\text{dog} \rightarrow \text{cog}$: inserts c, so optimal cost = 3
 - $\text{cog} \rightarrow \text{dog}$: inserts d, so optimal cost = 2
 - Therefore, the optimal costs are not symmetric.

Frequency:

1. No.
2. Actions are reversible, but cost depends on the resulting word: $1 + \text{rarity}(\text{result})$. Reversing an action changes the resulting word, so it can change the cost.
3. Counterexample:
 - $\text{rarity}(\text{dog}) = 5.819885$
 - $\text{rarity}(\text{cog}) = 11.393166$
 - $\text{dog} \rightarrow \text{cog}$: $1 + 11.393166 = 12.393166$
 - $\text{cog} \rightarrow \text{dog}$: $1 + 5.819885 = 6.819885$
 - Therefore, the optimal costs are not symmetric.

Question 4: LLM heuristic audit

1. Model, Prompt, Proposed heuristic

- **Model:**
 - Perplexity (default/Auto model selection)
- **Exact Prompt:**

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

- **Proposed Heuristic:**
 - $r_{\min}$ = minimum rarity among legal dictionary words of the same length
 - $h(n) = (1 + r_{\min}) \cdot d(n, g)$
 - where $d(n, g)$ is the number of positions in which state n differs from the goal, with $h(g) = 0$.

2. Precise Formula

- For a 3-letter problem:
 - $r_{\min} = 0.016226212$
- For a 4-letter problem:
 - $r_{\min} = 0.35150638$
- Thus:
 - $h(n) = (1 + r_{\min}) \cdot d(n, g)$

3. Admissibility

- Any state with $d(n, g)$ mismatches requires at least $d(n, g)$ actions, because one action changes only one position.
- Every frequency-cost action costs at least $1 + r_{\min}$.
- Therefore every solution from n costs at least:
 - $d(n, g) \cdot (1 + r_{\min})$
- Hence $h(n)$ never overestimates the true optimal remaining cost, so it is admissible.
- Because $r_{\min} > 0$ for both relevant word lengths, $h(n) > h_0(n)$ for every non-goal state.

4. Repair

- No repair was needed.
- The proposed heuristic is admissible and satisfies the assignment's definition of "more informative" than h_0 .

5. Experimental comparison

Problem	Heuristic	Solution cost	Nodes expanded
WHY $\rightarrow$ ASK	h_0	38.85540652	755
WHY $\rightarrow$ ASK	rarity-min	38.85540652	754
BEAR $\rightarrow$ DUCK	h_0	38.2402284	720
BEAR $\rightarrow$ DUCK	rarity-min	38.2402284	631

- Both heuristics returned the same optimal cost on both problems.
- The rarity-aware heuristic reduced expansions:
 - WHY $\rightarrow$ ASK: 755 $\rightarrow$ 754
 - BEAR $\rightarrow$ DUCK: 720 $\rightarrow$ 631
- The evidence therefore supports using the proposed heuristic, although the improvement is negligible on WHY $\rightarrow$ ASK and substantially larger on BEAR $\rightarrow$ DUCK.
 - These experiments support usefulness, but the admissibility claim comes from the proof, not from the tests.

Generative AI-use disclosure

I used ChatGPT (GPT-5.6 Sol) to help interpret the assignment requirements, assist with the implementation of `dc.py`, check my reasoning, and refine the presentation of my answers. I verified the implementation using the supplied tests, UCS/A* experiments, Google Sheets calculations, and additional Python checks.